

Epiphany's Body: A Proprioception Whitepaper

GameCult / EpiphanyAgent

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Abstract

Epiphany is an opinionated Codex-derived agent substrate built around a hard requirement: the model must model the thing it changes. This whitepaper maps Epiphany's **Body**: the repository, Rust crates, vendored bridge machinery, typed state stores, schema catalog, runtime binaries, protocol surfaces, operator artifacts, and external interface boundaries through which Epiphany senses and acts. The map is written from the Proprioception faculty: not as product copy, not as lore, but as an architecture overview whose owner, input, output, invariant, and cut lines remain inspectable.

1 Definition

In Epiphany doctrine, the Body is the substrate Epiphany acts through and senses: repository files, tools, runtime state, prompts, logs, interfaces, permissions, hosted systems, and editor/runtime bridges. Body is not Mind. Body is changed and sensed; Mind admits durable state. Proprioception is the faculty that models the Body: architecture, dataflow, seams, constraints, invariants, and the difference between the map and the substrate.

The practical definition in this repository is:

- the root checkout at `E:\Projects\EpiphanyAgent`;
- Epiphany Rust crates such as `epiphany-core`, `epiphany-state-model`, `epiphany-codex-bridge`, model/tool adapters, and OpenAI transport crates;
- vendored substrate in `vendor/codex`, `vendor/cultcache-rs`, `vendor/cultnet-rs`, and `vendor/cultmesh-rs`;
- typed state stores under `state/`;
- schemas under `schemas/cultnet/`;
- local operator scripts and generated witness artifacts;
- adjacent service/client bodies: Aquarium, Odin/Gjallar, Eve/CultUI, Bifrost, VoidBot, Rider, Unity, Discord, and future GameCult Verse hosts.

2 Executive Map

Epiphany's current Body is a hybrid machine in transition. Its center of gravity has moved into typed Rust/CultCache organs, but compatibility arteries still run through vendored Codex. The live target is not a cleaner pile of Codex routes. The target is an Epiphany-native CultCache/CultMesh/CultNet

body with a relatively vanilla Codex-derived reliquary retained for OpenAI subscription auth and model transport.

Body Region	Current Owner	Invariant
Durable thread state	<code>epiphany-state-model</code> , <code>epiphany-core</code>	State shape is typed; prompt rendering is a projection, not authority.
Domain policy	<code>epiphany-core</code>	Verdicts, launch policy, acceptance rules, runtime lifecycle, and mutation law live here, not in bridge DTOs.
Compatibility bridge	<code>epiphany-codex-bridge</code>	Translate Codex edge shapes; never become a second policy throne.
Vendored Codex	<code>vendor/codex</code>	Retain host/app-server/auth/model compatibility where useful; do not let Codex own Epiphany prompts, scheduler, or state.
Runtime spine	<code>epiphany-core/src/runtime_spine.rs</code> and <code>state/runtime-spine.cc</code>	Runtime identity, jobs, events, receipts, tool/model documents are typed CultCache records.
Organ memory	<code>state/agents.cc</code> ; agent-memory CLI	Resident work organs use lean work-organ state; Persona uses portable Persona state.
Heartbeat physiology	<code>state/agent-heartbeats.cc</code> via <code>epiphany-heartbeat-store</code>	Scheduling, cooldown, rumination, routine, cognition lanes, and active-turn freeze are scheduler physiology, not project truth.
Local Verse	CultMesh over <code>state/local-verse.cc</code>	Local Verse context is compact prompt/operator context; private state does not become public by flipping a flag.
Schema catalog	<code>schemas/cultnet/index.json</code>	JSON schema publishes contracts; typed Rust documents remain runtime authority.
Operator artifacts	<code>.epiphany-*</code> , <code>tools/epiphany_local_run.ps1</code>	Artifacts witness and lower state; they do not become state owners.

3 Repository Body

The checkout has ten primary Epiphany Rust crates:

- `epiphany-core`: the main domain organ. It owns state-update validation, launch documents, runtime spine, heartbeat, memory graph, Persona projection, Substrate Gate, Mind, Eyes, Hands, Soul, Continuity, CultMesh integration, surface derivation, smokes, and local operator binaries.

- **epiphany-state-model**: the durable `EpiphanyThreadState` type plus `render_epiphany_state`. This is the typed Mind surface that prompt fragments and views read.
- **epiphany-codex-bridge**: the translation layer between typed Epiphany documents and vendored Codex JSON-RPC/app-server shapes. Its own crate header states that it is a bridge, not an authority organ.
- **epiphany-model-adapter**: provider-neutral model request, stream event, receipt, and status documents.
- **epiphany-openai-adapter**: OpenAI-shaped request/event/receipt and status documents.
- **epiphany-openai-auth-spine**: Codex-derived auth boundary over `codex-login`. It preserves honest Codex-compatible credential and token behavior rather than impersonating it from scratch.
- **epiphany-openai-codex-spine**: model transport over vendored Codex client machinery.
- **epiphany-openai-runtime**: executable model runtime, including **epiphany-model-runtime**.
- **epiphany-tool-adapter**: typed tool capability, invocation intent, and invocation receipt documents.
- **epiphany-tool-codex-mcp-spine**: quarantined Codex MCP adapter. It consumes typed tool invocation intents and emits typed receipts; raw MCP JSON remains adapter-edge cargo.

The vendored region contains:

- **vendor/codex**: the Codex-derived host and compatibility substrate. It is tracked directly, not as a submodule.
- **vendor/cultcache-rs**, **vendor/cultnet-rs**, and **vendor/cultmesh-rs**: repo-contained Rust Cult substrate.

4 State Body

The **state/** directory is the local persistent body. The important stores are:

Canonical CultCache store files use the `.cc` extension. The local state body has been migrated to that naming directly; serialization details are implementation, not file authority.

Store	Role
<code>state/map.yaml</code>	Canonical slow project map: objective, constraints, invariants, dataflow, active subgoals, current status, guardrails, and rejected paths.
<code>state/ledgers.cc</code>	CultCache-shaped evidence and branch ledger. Future behavior-changing lessons go here, not in an endless activity feed.
<code>state/agents.cc</code>	Organ self-memory and Persona projection source. Resident work organs use <code>work_organ</code> ; public/person-shaped agents use <code>persona</code> .

<code>state/agent-heartbeats.cc</code>	Heartbeat initiative, cooldown, pending turns, cognition lanes, routine physiology, and scheduler pressure.
<code>state/runtime-spine.cc</code>	Runtime identity, sessions, jobs, job results, events, worker launch requests, role/reorient findings, receipts, model/tool documents, and schema catalog surfaces.
<code>state/thread-state.cc</code>	Native thread-state store.
<code>state/local-verse.cc</code>	Local CultMesh Verse context for compact operator and prompt-context packets.
<code>state/memory-graph.cc</code>	Local memory graph store.
<code>state/persona-discord.toml</code>	Persona Discord boundary configuration.
<code>state/void-memory.toml</code>	Void memory bridge configuration.
<code>state/scratch.md</code> and <code>state/scratch-compaction-*.md</code>	Scratch surfaces. Scratch is a workbench, not canonical memory.

Lock files sit beside several stores. Some local generated state is currently untracked, including `state/local-verse.cc`, `state/memory-graph.cc`, and `.epiphany-character-loop/`.

5 CultNet Schema Body

The schema catalog in `schemas/cultnet/index.json` is the public nerve chart. Current local grouping from the catalog is approximately:

- 10 agent-state schemas;
- 5 heartbeat schemas;
- 21 intent schemas;
- 8 model/OpenAI schemas;
- 3 tool schemas;
- 6 runtime schemas;
- 17 surface schemas;
- 4 receipt/ledger schemas;
- 7 other schemas.

The catalog includes runtime identity/session/job/event/result documents, model requests and receipts, tool invocation intents and receipts, Persona and work-organ state, heartbeat state, operator CultMesh documents, scene/freshness /context/graph/pressure/reorient/roles/coordinator surfaces, Unity and Rider bridge surfaces, Void memory, and repo-initialization documents.

The invariant is simple: schema JSON describes the contract. Load-bearing runtime state should be typed Rust documents persisted through CultCache or projected through CultMesh/CultNet. JSON may remain at hostile edges, schema publication, sealed forensic artifacts, or named quarantine experiments.

6 Runtime Spine

`epiphany-core/src/runtime_spine.rs` is one of the main spinal cords of the Body. It imports and registers:

- runtime identity, session, job, job-result, and event records;
- worker launch requests and role/reorient worker result documents;
- model adapter, OpenAI adapter, and tool adapter documents;
- surface document types for scene, freshness, context, graph query, pressure, reorient, CRRC, jobs, roles, role result, reorient result, planning, coordinator, Persona, Void memory, repo initialization, repo birth runner, Rider bridge, and Unity bridge;
- organ receipts for Mind, Eyes, Hands, Soul, Continuity, and Substrate Gate;
- CultMesh operator status, snapshots, run intents, run receipts, Verse policy, global room policy, and organ contract documents.

Runtime state is machinery telemetry. It is not project truth. Durable thread state may link to runtime records through runtime links, but it must not mirror or duplicate their lifecycle fields as a competing owner.

7 Faculties And Protocol Machinery

The embodied sub-agents are:

- **Self**: coordination, routing, authority boundaries.
- **Persona**: public and social expression.
- **Imagination**: futures, plans, objective drafts, projection.
- **Eyes**: research, source inspection, evidence ingress.
- **Proprioception**: internal model of Body shape, dataflow, architecture, checkpoints, and seams.
- **Hands**: commands, edits, commits, rollbacks, external action.
- **Soul**: verification, invariants, ethics, falsification.

The machinery surfaces are:

- **Body**: substrate. It is sensed and changed; it does not decide.
- **Mind**: persistent state and state admission.
- **Continuity**: compaction, recovery, sleep, stale-turn repair, and reorientation receipts.
- **Substrate Gate**: scoped repository/substrate access.

Every embodied sub-agent depends on the other embodied sub-agents, but dependency does not collapse ownership. A Persona turn, for example, should move through Substrate-Gate-scoped facts, Mind state, social stimulus, organ dependencies, Imagination as Projector, Persona as natural-language actor, and Mind as Interpreter. None of those steps grants Persona hidden state authority.

8 Bridge And Codex Reliquary

`epiphany-codex-bridge` states its boundary plainly: it translates Codex `ThreadEpiphany*` DTOs into Epiphany core documents, projects typed Epiphany surfaces back into legacy Codex responses, and calls narrow host traits while Codex still owns some thread persistence. It must not own verdicts, launch policy, reorientation policy, acceptance rules, runtime-spine lifecycle, or durable invariants.

The retained Codex body currently matters for:

- Codex-compatible auth identity and token refresh through `codex-login`;
- OpenAI/Codex model transport through vendored client machinery;
- useful app-server and streaming compatibility affordances while Epiphany surfaces are still evacuating to native CultNet/CultMesh contracts.

The starvation doctrine is the deletion line: Codex may be a bridge and a reliquary, but not Epiphany's prompt authority, state owner, scheduler, runtime truth, operator API owner, or plugin marketplace brain.

9 Operator And Interface Body

Current operator access is partly native and partly witness-artifact based:

- `tools/epiphany_local_run.ps1` is the one-command local operator wrapper.
- Native binaries such as `epiphany-mvp-status`, `epiphany-mvp-coordinator`, `epiphany-verse-query`, `epiphany-operator-run`, `epiphany-operator-snapshot`, `epiphany-eve-surface`, `epiphany-runtime-spine`, `epiphany-heartbeat-store`, and `epiphany-agent-memory-store` inspect or update typed state.
- Local artifact directories such as `.epiphany-run/`, `.epiphany-dogfood/`, `.epiphany-smoke/`, `.epiphany-rumination/`, `.epiphany-gui/`, and `.epiphany-character-loop/` witness work and lower typed state for operator inspection.
- Aquarium is the richer adjacent operator client. It should render and steer typed state, not become a state owner.
- Eve/CultUI is the intended GUI/TUI projection mechanism for typed CultMesh surfaces.
- Odin/Gjallar/Yggdrasil are the intended GameCult Verse discovery and hosting organs for daemon affordances and important trusted Verses.

The service contract invariant is that status surfaces, dashboards, shell wrappers, bridge adapters, app-server compatibility layers, and renderers must read typed projections or emit typed command intents. They must not directly mutate Epiphany truth.

10 Primary Dataflows

10.1 State Rehydration And Prompt Projection

Durable thread state lives as typed `EpiphanyThreadState`. On re-entry, state is reconstructed from rollout/native state, then rendered into bounded prompt-facing context through `render_epiphany_state`. Prompt injection is read-only. If a model learns something from that prompt, admitting it back into durable state requires an explicit update, proposal, promotion, role acceptance, or other reviewed write path.

10.2 View And Surface Projection

Scene, jobs, freshness, context, graph query, planning, pressure, reorientation, CRRC, roles, role result, reorient result, coordinator, Persona, bridge, and environment surfaces are projections. Their job is to make typed state visible and actionable. A projection does not own its inputs. A dashboard is a window, not the vault.

10.3 Worker Launch And Runtime Results

Launch requests are typed documents. Runtime spine opens jobs, links them to durable bindings, records model/tool activity, and stores results/receipts. Role and reorientation findings remain reviewable until Mind and relevant gates accept them. Implementation now routes through Hands-shaped branch-turn constraints and should yield one branch/commit evidence slice before Proprioception refresh.

10.4 Tool Calls

Tool use travels through typed capability, invocation intent, and invocation receipt documents. The Codex MCP adapter is an external compatibility executor. MCP's raw JSON belongs at that boundary; Epiphany's internal tool state is the typed intent/receipt pair.

10.5 Local Verse And Daemon Awareness

Local Verse context is queried through CultMesh-backed state and can import Gjallar daemon affordance summaries. The desired daemon path is Odin/Gjallar for discovery, Eve/CultUI for inspection surfaces, CultMesh/CultNet for state and commands, and provider-owned receipts for actions. MCP is not the native GameCult daemon-control path.

11 Invariants

1. CultCache documents are the data model.
2. CultNet is the wire for Epiphany-controlled subsystems.
3. CultMesh is the preferred ergonomic local Verse and operator-store layer over raw CultCache/CultNet.
4. JSON schema may publish the contract; generic JSON must not become load-bearing internal state when both sides are Epiphany-owned.

5. Codex is retained for honest compatibility, especially OpenAI auth and model transport. It must not become Epiphany's state/process/prompt owner.
6. Runtime state is machinery telemetry, not project truth.
7. Operator artifacts are witnesses, not owners.
8. Persona is public/person-shaped state; resident work organs use lean work-organ state.
9. Heartbeat is scheduler physiology, not project truth.
10. Substrate Gate grants access; Eyes inspects evidence; Hands acts; Soul verifies; Mind admits durable state; Self routes; Proprioception maps Body.

12 Current Cut Line

The Body is coherent enough to inspect and extend. The next cuts are concrete:

- Keep `thread/epiphany/*` JSON-RPC as compatibility projection only. New operator authority belongs in typed CultNet/CultMesh contracts.
- `codex_message_processor.rs` still participates in routing and response projection; move Epiphany-owned routing out through the bridge and native runtime surfaces.
- Some `serde_json::Value` and JSON payloads survive at ingress, model output, schema, sealed artifacts, and quarantine seams. Preserve them only at those boundaries; typed documents own Epiphany-internal state.
- Publish Eve/CultUI compositions from provider-owned CultMesh state; renderers lower the composition graph without becoming state owners.
- Treat local Verse and memory graph stores as generated runtime state. They are inspected through tools and receipts, not tracked as repository canon.
- The persisted next action after this pass remains the non-ephemeral Hands branch-turn smoke and Proprioception refresh proof.

13 Conclusion

Epiphany's Body is no longer a prompt style pasted onto Codex. It is a growing typed substrate: Rust domain organs, CultCache stores, CultMesh local Verse documents, CultNet schema contracts, runtime-spine receipts, heartbeat physiology, memory graph state, model/tool adapter documents, and local operator surfaces. The current cut line is split authority: Codex host routes, JSON-RPC compatibility, JSON cargo, and operator artifacts must remain compatibility or witness surfaces rather than organs.

The cut line is therefore architectural, not cosmetic. Keep Codex as a narrow auth/model/compatibility reliquary. Move Epiphany-owned state, scheduler policy, prompt authority, worker lifecycle, operator surfaces, daemon discovery, and action receipts through typed CultCache, CultMesh, CultNet, and Eve/CultUI. That is the Body becoming legible enough to be trusted and sharp enough to keep building.